Homework 4

Due: <u>November 21, 2025</u>

Total Points: 100

Instructions:

- Show all reasoning or calculations where relevant.
- When asked for an interpretation, explain your answer clearly in words.
- You may use R for any applied components. Include all necessary R code and short interpretations.

Q1. (20 points) You are studying a dataset of 500 patients, each with 200 biomarkers measured at baseline. You plan to visualize the relationships between individuals using a low-dimensional embedding (2D or 3D). Consider the following situations and state whether a **global** or **local** dimension reduction method would be more appropriate. Justify your answer in 2–3 sentences.

(a) You believe there is a single underlying continuous "disease severity" score that explains most of the biomarker variation.

(b) You are interested in identifying small subgroups of patients with very similar biomarker profiles but distinct from the rest of the population.

(c) You plan to create a 2D visualization of single-cell RNA expression data to identify cell-type clusters.

(d) You have time-series physiological data (e.g., heart rate, sleep, and activity) and want to summarize the overall level and variability across participants.

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Q2. (20 points) Complete the table below by summarizing the key ideas for each dimension reduction method.

Method	Main Objective	When to Use / Key Characteristics
PCA		
Factor Analysis		
ICA		
t-SNE		
UMAP		

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Q3. (30 points) In this exercise, you will analyze country-level global health indicators from the `gapminder` dataset. This dataset contains country-level statistics including GDP per capita, life expectancy, and population size from 1952–2007.

Step 1: Load the data.

```
# Install/load packages
install.packages("gapminder")
library(gapminder)
data(gapminder)

# Subset the most recent year
gap <- subset(gapminder, year == 2007)

# Keep quantitative variables only
gap_sub <- gap[, c("lifeExp", "gdpPercap", "pop")]
rownames(gap_sub) <- gap$country
```

Step 2: Dimension Reduction Tasks

(a) Perform a **Principal Component Analysis (PCA)** on the standardized variables. Report the proportion of variance explained by each component and create a biplot showing countries colored by continent.

(b) Conduct a **Factor Analysis** with two latent factors. Interpret what each factor appears to represent in terms of economic and health dimensions.

(c) Perform an **Independent Component Analysis (ICA)** with two components. Compare the ICA components to the PCA loadings and discuss differences in interpretation.

(d) Apply both **t-SNE** and **UMAP** with two dimensions. For each approach, create a scatterplot for each method with points colored by continent. UMAP is supposed to represent more global vs. local structure than t-SNE. Do the results appear to back this statement up? Why?

(e) Include one paragraph comparing PCA, FA, ICA, t-SNE, and UMAP for this dataset.

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Q4. Conceptual and Comparative (15 points)

Complete the table below comparing the following clustering methods.

Method	**Clustering Criterion / Model**	**Key Strengths and Limitations**
K-Means		
Gaussian Mixture Model (GMM)		
PAM (Partitioning Around Medoids)		
Hierarchical Clustering		

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Q5. Application: Violent Crime Rates by US State (15 points)

You will cluster crime data using publicly available data on multiple pollutants.

Step 1: Load the data. This dataset is already in R. You can look at it with:

```
head(USArrests)
```

Step 2: Clustering Tasks

(a) Apply **K-Means** clustering to the states using the four crime measures. Determine the optimal number of clusters using the `fviz_nbclust()` function from `factoextra`.

(b) Fit a **Gaussian Mixture Model (GMM)** to the same data and compare the BIC for models with 2–6 clusters.

(c) Apply **PAM (Partitioning Around Medoids)** and **Hierarchical Clustering** using Euclidean distance. Create and briefly interpret a dendrogram.

(d) Compare the clustering results across methods. Which sites consistently cluster together?

(e) How could the clustering results inform violent crime policy?